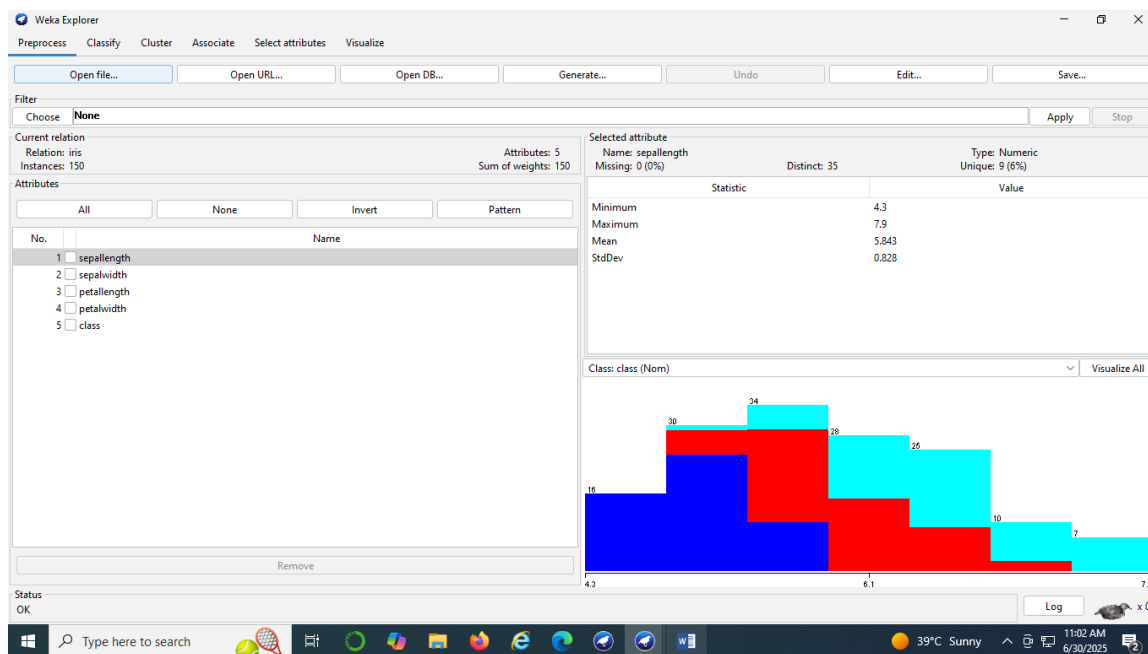
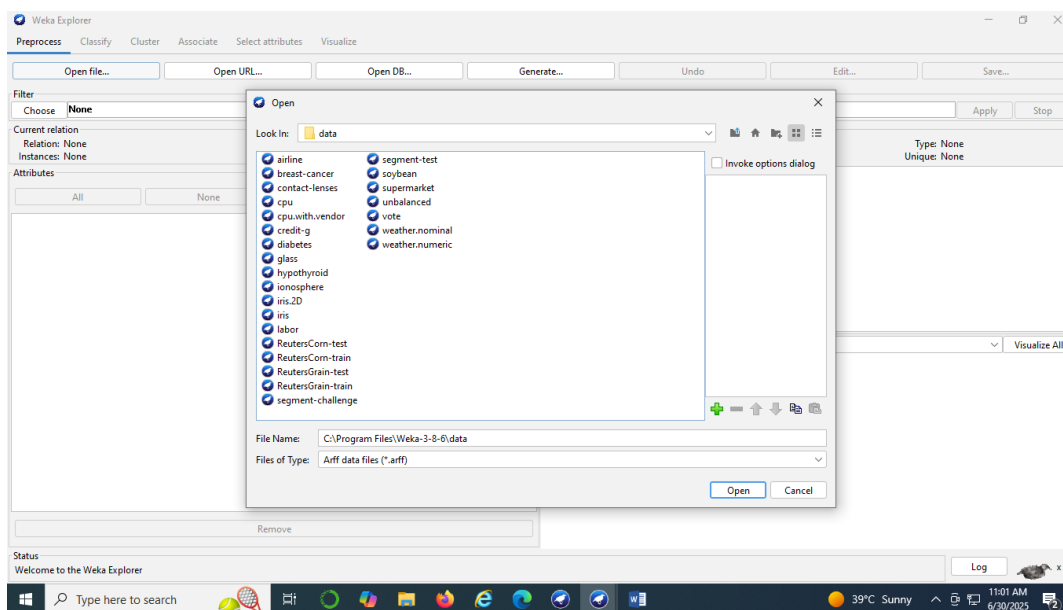


Exercise 10: Demonstrate two case studies using Weka Tool.

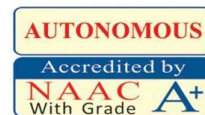
Case study 1: ZeroR





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Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier

weka

- classifiers
 - bayes
 - functions
 - lazy
 - meta
 - misc
 - rules
 - DecisionTable
 - JRip
 - MSRules
 - OneR
 - PART
 - ZeroR**
 - trees

Close

Status OK

Type here to search

39°C Sunny 11:03 AM 6/30/2025

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier

Choose **ZeroR**

Test options

- ☐ Use training set
- ☐ Supplied test set
- ☒ Cross-validation Folds 10
- ☐ Percentage split % 66

More options...

(Nom) class

Start Stop

Result list (right-click for options)

11:03:30 - rules.ZeroR

Classifier output

=== Run information ===

Scheme: weka.classifiers.rules.ZeroR
Relation: iris
Instances: 150
Attributes: 5
sepalength
sepalwidth
petalength
petalwidth
class

Test mode: 10-fold cross-validation

=== Classifier model (full training set) ===

ZeroR predicts class value: Iris-setosa

Time taken to build model: 0 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	50	33.3333 %
Incorrectly Classified Instances	100	66.6667 %
Kappa statistic	0	
Mean absolute error	0.4444	
Root mean squared error	0.4714	
Relative absolute error	100 %	
Root relative squared error	100 %	
Total Number of Instances	150	

=== Detailed Accuracy By Class ===

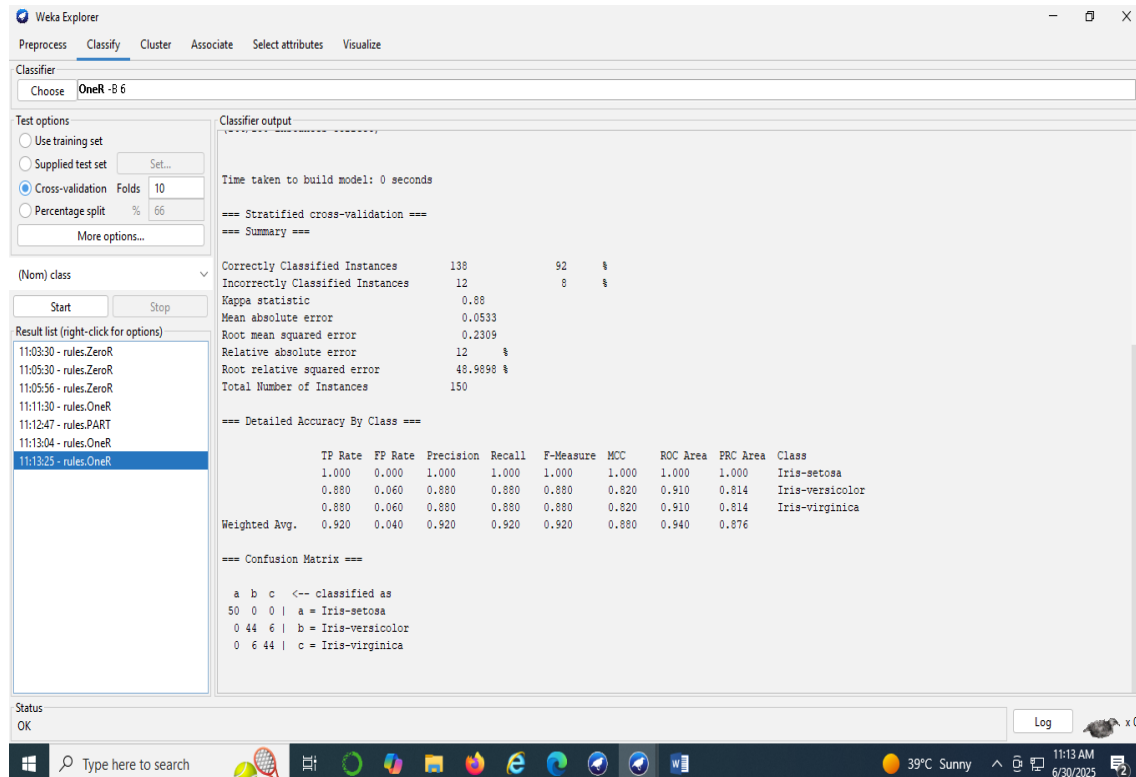
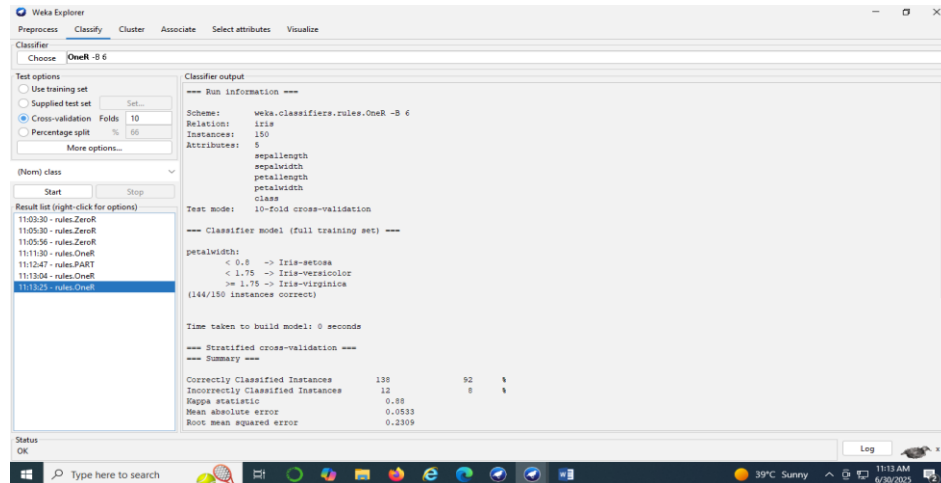
Status OK

Type here to search

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Case study2:

OneR short for "**One Rule**", is a simple, yet accurate, classification algorithm that generates one rule for each predictor in the data, then selects the rule with the smallest total error as its "one rule"





Case study 3:

PART:

Part classification is a systematic approach which involves organizing parts into categories and subcategories based on specific attributes. This can include design characteristics, manufacturing processes, reducing costs, and functional similarities. Effective classification helps in avoiding duplicate parts, streamlining inventory management, and enhancing the retrieval of existing parts for reuse in new designs.

The screenshot shows the Weka Explorer interface with the PART classifier selected. The 'Test options' section shows 'Cross-validation' with 'Folds' set to 10. The 'Classifier output' section displays the following results:

Number of Rules : 3
Time taken to build model: 0.01 seconds

=== Stratified cross-validation ===
=== Summary ===

Metric	Value
Correctly Classified Instances	141
Incorrectly Classified Instances	9
Kappa statistic	0.91
Mean absolute error	0.0482
Root mean squared error	0.1794
Relative absolute error	10.8379 %
Root relative squared error	38.0567 %
Total Number of Instances	150

=== Detailed Accuracy By Class ===

TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
0.980	0.000	1.000	0.980	0.990	0.985	0.990	0.987	Iris-setosa
0.940	0.060	0.887	0.940	0.913	0.868	0.954	0.878	Iris-versicolor
0.900	0.030	0.938	0.900	0.918	0.879	0.959	0.914	Iris-virginica
Weighted Avg.								
0.940	0.030	0.941	0.940	0.940	0.911	0.968	0.926	

=== Confusion Matrix ===

```
a b c <-- classified as
49 1 0 | a = Iris-setosa
0 47 3 | b = Iris-versicolor
0 5 45 | c = Iris-virginica
```

By Observing the three classifiers like ZeroR, OneR, PART. The PART classifier gives the best Classifier gives the Highest accuracy results.